Decision Assistance and Steering in Health Insurance

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Abstract

In many markets, consumers rely on an agent or some other intermediary to aid in decision making. The influence of such agents on consumer decisions, and the firm's influence on the behaviors of its agents, have important implications for the efficiency of these markets. In this paper, we study the welfare effects of decision assistance and firm steering in health insurance using the population of enrollments from the California Affordable Care Act (ACA) exchange from 2014 to 2019. We find that enrollees are 22% more likely to select a silver plan when using some form of decision assistance and 42% less likely to select a "dominated" plan. We also find that firms have considerable ability to steer consumers' decisions using agent commissions, where a $1 increase in monthly commissions paid to agents shifts broker-assisted enrollment as much as a $3.29 reduction in net monthly premiums. Embedding these estimates in a structural model of insurer pricing, we find that the welfare value of assistance comes almost entirely from keeping households insured rather than from improving which plan they choose, and that the distortion from commissions is small by comparison.

1 Introduction

Health insurance markets are unique in many respects, not least of which is the increasing complexity of choosing an optimal health insurance plan. Such complexity has been well-documented in studies of health insurance choice in Medicare (Abaluck and Gruber, 2011; Ketcham et al., 2012; Gruber, 2017). One way to reduce the burden of this complexity is to provide professional decision support through private insurance agents or public assistance programs, both of which are available in the California health insurance exchanges created under the Affordable Care Act (ACA). Such decision assistance, however, also introduces the potential for private insurers to steer enrollees into potentially suboptimal plans. Commissions paid to facilitate any steering may also increase plan premiums for all enrollees. In this paper, we examine the role of decision assistance on health insurance plan choice and its implications for consumer welfare.

Our analysis is based on household enrollment data from the California ACA exchange. Our final data consist of nearly 8 million household-year observations from 2014 to 2019. The data include a variety of household characteristics, plan choices and plan characteristics, and information on the type of decision assistance used by the enrollee (if any). Using these data, we can precisely identify each household's choice set, the premium paid for each plan in the choice set, and any premium and cost sharing subsidies for which the household is eligible.

Our empirical analysis proceeds as follows. First, we examine whether decision assistance affects plan choice. Our estimand of interest is the average treatment effect on the treated (ATT) for enrollees receiving decision assistance. We estimate an inverse propensity weighted conditional logit regression on unassisted households and use it to predict the plans that assisted households would have chosen absent assistance, forming the ATT as the difference between observed and predicted enrollment, as discussed in more detail in Section 4. We estimate that assisted households are 12.3 percentage points more likely to select a silver plan, roughly 22% above the counterfactual silver share, the majority of which comes from a reduction in the probability of selecting a bronze plan.

Second, we consider the potential for decision assistance to "improve" plan choice. While identifying "improvement" is not generally feasible in our data, we can identify specific instances in which the observed plan choice is dominated by some other plan in an individual's choice set. For example, gold and platinum tier plans are dominated for any household that is eligible for cost-sharing subsidies and has income below 150% of the federal poverty level. We can therefore identify dominated choices and examine whether those with decision assistance appear less likely to select a dominated plan. As with our initial model of plan choice, our estimand is the ATT for households receiving decision assistance, and our estimator is an inverse propensity weighted logit regression fit on unassisted households and used to predict the counterfactual for assisted households.

Our results suggest that assisted households are 1.6 percentage points less likely to make a dominated choice (about a 42% relative reduction). We also find evidence of heterogeneity in the effects of different forms of decision assistance. In particular, we find that households using publicly provided decision assistance are 0.9 percentage points less likely to select a dominated plan (about a 25% relative reduction), while households using a private insurance agent are 1.8 percentage points less likely to select a dominated plan (about a 48% relative reduction).

Third, we build a structural model of the exchange that pairs a nested logit model of household plan choice with a model of insurer premium-setting, embedding decision assistance and the commissions insurers pay their agents in both. Households choose among plans with the help of a navigator or a commissioned agent, while insurers set premiums through a pricing first-order condition that accounts for risk adjustment, subsidies, and the commissions they pay to steer broker-assisted enrollment. Estimating both sides on California exchange data lets us separate the value that assistance provides from the steering that commissions induce. Insurers recover part of the commissions they pay through the premiums they charge, so this steering may raise premiums for all enrollees, including the unassisted who never consult an agent. Because premiums and commissions are set jointly rather than taken as given, our model traces this spillover onto the broader market. We estimate that commissions meaningfully steer the enrollment of broker-assisted households. Evaluated at the mean price sensitivity of those households, a $1 increase in monthly commissions paid to private insurance agents shifts their enrollment as much as a $3.29 reduction in net monthly premiums, and in percentage terms a one percent change in commissions moves broker-assisted demand roughly two-fifths as much as a one percent change in premiums.

Finally, we use the estimated model to evaluate counterfactual policies that fall into three general categories: 1) withdrawing decision assistance; 2) reshaping the commission schedule while leaving assistance in place; and 3) substituting public navigators for private agents. Withdrawing assistance has by far the largest effect, raising the uninsured share by 7.7 percentage points, whereas reshaping commissions moves welfare by at most a few tens of dollars per member per year and leaves coverage essentially unchanged, and substituting one channel for the other changes coverage only modestly. Decomposing the welfare change confirms that it operates almost entirely through keeping households insured, and the risk protection that coverage provides, rather than through steering households toward better plans within the market. Commissions do create a spillover, since insurers pass part of the commissions they pay into the premiums every enrollee faces, including the unassisted, but this spillover and the steering distortion are both small once separated from the value of the assistance channel, in part because standardized benefit designs make steering across plans within a metal tier a low-stakes margin relative to the decision of whether to enroll.

Taken together, these findings contribute to three distinct literatures. First, our examination of decision assistance and health insurance relates to the literature on the role of experts in decision making.[1] Theoretical models of the role of experts and advice in a principal-agent framework include Szalay (2005), Alonso and Matouschek (2008), and Armstrong and Vickers (2010). In an experimental setting, Schotter (2003) finds that individuals are much more likely to follow the advice of an expert and that such behavior is welfare-improving. In education, Borghans et al. (2015) find large effects of high school counselors on students' college field choices, and Bettinger et al. (2012) estimate large effects of personal assistance on the probability of applying for and receiving college financial aid, with subsequently large effects on college attendance. Finkelstein and Notowidigdo (2019) similarly find large effects of both improved information and personal assistance on the probability of enrollment in the Supplemental Nutrition Assistance Program among eligible households.

A related literature considers the effects of recommendations, sometimes generated algorithmically, on decisions. Bundorf et al. (2024), for example, show that offering additional

[1]A broader literature considers "persuasion", which includes the role of advertising and media reports. We are interested in the more narrow literature specifically dealing with decision assistance in the form of third-party "experts" or advice. See DellaVigna and Gentzkow (2010) for a general survey of the empirical evidence on persuasion.

information alongside an algorithmic expert recommendation significantly affects an individual's choice of prescription drug plans. Kling et al. (2012) also estimate large effects of personalized cost information on plan switching in the Medicare Part D market. To our knowledge, the literature on experts and decision making in the context of health insurance focuses on unbiased and often anonymous "experts" (e.g., an algorithm suggesting the best plan for an individual). Our setting is unique in that individuals have access to a public "navigator," who receives no compensation for any plan choice, and a private insurance agent who receives a commission that varies by plan.

Second, we contribute to the literature examining the prevalence and underlying mechanisms of poor decision making in health insurance. In the employer-provided insurance market, Liu and Sydnor (2021) document that over half of employers offer a dominated plan among their menu of insurance options available to their employees, and Sinaiko and Hirth (2011) and Bhargava et al. (2017) find that a large share of employees regularly select such dominated insurance plans. Abaluck and Gruber (2011), Ketcham et al. (2012), Zhou and Zhang (2012), Heiss et al. (2013), Gruber (2017), Ho et al. (2017) and many others similarly examine poor decisions when selecting a Medicare prescription drug (Part D) plan. The consensus from this literature is that individuals regularly make *ex ante* poor health insurance decisions, often selecting plans that are either financially dominated by other plan options or most likely suboptimal given the enrollee's existing medical conditions. While not definitive, a common theme in this literature is that poor health insurance decisions are due in part to an inability or unwillingness of enrollees to accurately evaluate their plan options (Bhargava et al., 2017; Hanoch et al., 2009).

A few recent studies consider poor health insurance decisions in the context of the ACA and the role of public navigators.[2] For example, Orzol and Hula (2018) find that "Enroll America," a national non-profit organization designed to expand health insurance enrollment in the U.S., significantly increased enrollment in the ACA exchange plans. Myerson and Li (2022) also find that federally funded assistance programs significantly increased the probability of enrollment in Medicaid, although they find little effect of such federal funding

[2]Ketcham et al. (2012) find that observed poor health insurance decision making will tend to improve over time, thereby partially alleviating the need for any explicit intervention; however, unlike Medicare Advantage or Medicare Part D, the majority of enrollees in California exchange plans are not continuously enrolled over multiple years. Therefore, there exists relatively little opportunity for enrollees to sufficiently identify poor decisions and update plan choices accordingly.

on enrollment in exchange plans. Sommers et al. (2015) similarly show that access to navigators is highly correlated with Medicaid enrollment, mirroring earlier findings from Aizer (2003) in her study of Medicaid enrollment and application assistants in California. Aizawa and Kim (2025) distinguish between "public" provision (via government advertising) versus private provision (via insurer advertising) on plan choice, where they find that both forms of advertising affect participation in the market and that private advertising has a particularly large effect on the intensive margin of plan choice; however, the welfare effects of private advertising are unclear due to business stealing and costs of advertising.

Our contribution to this literature is most closely related to Aizawa and Kim (2025) in that we consider both private and public decision assistance; however, we examine this in the context of one-on-one assistance rather than general exposure to advertising. We also consider the potential for private agents to improve health insurance decisions while allowing for potential bias introduced through commissions (i.e., steering), and we compare outcomes from the use of private agents to those of public navigators. Our analysis envisions private agents as helping to resolve an information problem, the costs of which include potential steering into suboptimal plans and spillover effects of commission payments onto premiums. Whether the costs of such private agents exceed the benefits or are best borne by public agents without steering incentives is an empirical question which we pursue in Section 5.

Third, we contribute to the structural literature on insurer pricing in regulated health insurance exchanges. A series of studies models consumer plan choice jointly with insurer pricing, including Starc (2014) in Medigap and Tebaldi (2025) and Saltzman (2019, 2021) in the ACA exchanges, on whose framework our supply side builds. This literature studies how premiums, subsidies, and risk adjustment shape both the extensive margin of enrollment and the intensive margin of plan generosity, a distinction Saltzman (2021) frames as underenrollment versus underinsurance. We extend it by introducing decision assistance and the commissions insurers pay the agents who provide it, so that firms influence choices not only through premiums but through the recommendations their commissions steer. In doing so we connect this literature to work on conflicts of interest in financial advice and intermediation (Egan et al., 2019), where the compensation of an agent can diverge from the interests of the client. Our central welfare result speaks to the underenrollment-underinsurance distinction directly, since the value of assistance falls almost entirely on the enrollment margin rather than on the plan households choose.

Our findings speak directly to an unsettled policy question, namely who should provide decision assistance and how it should be paid for. Federal navigator funding has been cut, restored, and cut again across recent administrations. We find that shifting households from public navigators to private agents preserves most of the extensive-margin benefit of decision assistance, keeping coverage roughly intact, while introducing the modest steering that commissions create. Commissions also raise the premiums all enrollees pay, though we find this spillover to be small. The same concern is active in Medicare Advantage, where, although the rule was ultimately paused, CMS has attempted to cap broker compensation on the grounds that commissions steer beneficiaries toward plans that serve the broker's interest rather than their own.[3] Our estimates suggest the magnitudes at stake are small and may not warrant such policy concern.[4]

The remainder of the paper proceeds as follows. Section 2 describes the California ACA marketplace and the decision-assistance channels available to enrollees. Section 3 describes our data. Section 4 presents the design-based analysis of how decision assistance affects plan choice, including the probability of selecting a dominated plan. Section 5 develops and estimates the structural model of household plan choice and insurer pricing with commission-based steering. Section 6 uses the estimated model to evaluate the welfare effects of decision assistance and counterfactual policies that reshape commissions and substitute navigators for agents. Section 7 concludes.

2 Background and Institutional Details

One of the pillars of the Affordable Care Act (ACA) was the creation of regulated state insurance exchanges in which insurers sell private insurance plans directly to consumers. While all state exchanges must adhere to a minimum set of federal regulations, states vary in how they manage and regulate their exchanges. Since our analysis uses data from the California exchange, we focus on the institutional details of the California ACA marketplace.

[3]Centers for Medicare & Medicaid Services, *Contract Year 2025 Medicare Advantage and Part D Final Rule*, 89 Fed. Reg. 30448 (April 23, 2024).

[4]The Medicare Advantage setting differs from ours in that the extensive margin is narrower, since beneficiaries retain traditional Medicare regardless. Even at our low valuation of the cost of being uninsured, where that margin contributes least, reshaping the commission schedule barely moves welfare.

There are three general aspects of the California insurance exchange that are particularly relevant for our analysis: 1) enrollee choice sets and the types of plans available; 2) variation in premiums across individuals on the exchange; and 3) the presence of navigators and brokers to assist individuals when considering an exchange plan. We discuss the institutional details of each of these areas in more detail throughout the remainder of this section.

2.1 Choice Sets and Plan Types

Exchange plans are classified by a "metal" tier intended to easily summarize the actuarial value (AV) of a plan. Bronze plans reflect an AV of 60%, silver plans an AV of 70%, gold plans an AV of 80%, and platinum plans an AV of 90%. For plans on the California exchange, other cost-sharing parameters are standardized within a metal tier, such that the enrollee's financial burden across plans within a given metal tier is relatively homogeneous.[5] Each state also defines geographic rating areas, usually composed of counties, in which an insurer's premiums must be the same for consumers of the same age.

Most households face the same choice set based on their region and year of enrollment; however, some individuals (mainly those under age 30) are also eligible for basic catastrophic coverage. Insurers can also choose to serve only a part of a rating area. Such "partial" rating area coverage is relatively uncommon in California, and Fang and Ko (2025) show that insurers' decisions to enter a subset of counties in a rating area appear to be determined by entry costs and risk pools rather than strategic competitive behavior.

2.2 Premiums and Cost-sharing

Premium variation within a plan is more tightly regulated in California than in other state exchanges, with insurer price discrimination only allowed based on an enrollee's age and geographic residence. For example, insurers can charge a 64-year-old up to 3 times as much as a 21-year-old according to the default age rating curve.

The ACA also provides premium and cost-sharing subsidies to eligible enrollees.[6] The

[5] In other states, insurers can generally design plans with different cost-sharing provisions within a metal tier, but the regulations on the California exchange are more stringent.

[6] Premium subsidies are available to consumers who meet the following criteria: 1) income between 100 and 400% of the federal poverty level (FPL); 2) citizenship or legal resident status; 3) ineligible for public insurance such as Medicare or Medicaid; and 4) lack access to an "affordable plan offer" through employer-

premium subsidy is set to the difference between: 1) the premium of the benchmark plan, defined as the second-cheapest silver plan available to a given enrollee; and 2) the enrollee's income contribution cap, defined as the maximum percent of a person's income spent on premiums. Consumers can apply the premium subsidy toward the premium of any metal plan, whereas cost-sharing subsidies only apply to the purchase of a silver tier plan.

2.3 Decision Assistance

There are two forms of formal decision assistance available to potential enrollees on the California exchange: "navigators" and private insurance agents or brokers. For states participating in the federal exchange, navigators are mandated under the ACA and were initially federally funded, although funding has fluctuated substantially across administrations.[7] Most state-based exchanges, including California, offer similar navigator assistance to potential enrollees.

A navigator is available to help potential enrollees better understand their insurance options, and navigators receive no compensation tied to any specific plan choice. Navigators have a particularly large effect on Medicaid enrollment (Myerson and Li, 2022; Sommers et al., 2015; Aizer, 2003). Relatedly, navigators tend to work with households that have lower incomes, are currently uninsured, or include non-native English speakers. Among individuals receiving some form of assistance in our data, about 30% of households used a navigator.

Potential enrollees on the California exchange may also use a traditional private insurance agent or broker (hereafter, agents). An agent receives a commission only from the insurers with which the agent holds a contract, so an agent's compensation is tied to a subset of

sponsored insurance either as an employee or as a dependent. Cost-sharing subsidies are available to anyone with income $\leq$ 250% of FPL and serve to increase the actuarial value of a plan from 70% up to 94% (income $<$ 150% of FPL), 87% (income between 150% and 200% of FPL), and 73% (incomes between 200% and 250% of FPL).

[7] Federal navigator funding fell from \$63 million for the 2017 plan year to \$37 million for 2018 and \$10 million for 2019, and federal marketplace advertising was cut over the same period from \$100 million to \$10 million (U.S. Government Accountability Office, *Health Insurance Exchanges: HHS Should Enhance Its Management of Open Enrollment Performance*, GAO-18-565, 2018, pp. 23–24). The 2019 level is about 84 percent below the 2017 peak. Funding was later restored, reaching \$100 million for the 2025 open enrollment period, and reduced again to \$10 million in a February 2025 announcement effective for plan year 2026 (Centers for Medicare and Medicaid Services, press releases, August 26, 2024 and February 14, 2025). California, as a state-based exchange, funds its own navigator and outreach program through an assessment on plan premiums and was not directly affected by these federal changes (Covered California, *Fiscal Year 2017–18 Proposed Budget*, 2017, pp. 10, 17).

carriers rather than to every plan on the exchange. Covered California nonetheless certifies participating agents and requires them to present all available plans to a consumer without regard to the commission each carries, so an agent's stated obligation spans the full choice set even though only the contracted carriers pay that agent.[8] Agents are not restricted to any specific geography or region and can enroll individuals in the relevant rating area regardless of the physical office in which the agent resides. Among individuals receiving some form of assistance in our data, 70% used a private insurance agent.

Unlike navigators, agents are paid commissions for each enrollment into an exchange plan. The amount of commission varies by insurer and is generally set as a flat fee per enrollee or as a percentage of the premium paid by that enrollee. For new enrollees, flat-fee commissions typically range between \$100 and \$200, with percent commissions ranging between 2% and 5%. Figure 1 presents the observed commission rates by insurer over time, separately among insurers offering a flat-fee commission versus those offering a percentage commission. As is evident from the figure, there exists meaningful variation in commissions across insurers, although commission rates within a given insurer are largely stable over time.[9]

3 Data

Our analysis draws on several data sources, which we describe in turn before discussing sample construction and summary statistics.

[8] The requirement to present all plans is set out in Covered California's agency agreement, which directs certified agents to fairly and accurately present all available enrollment options and prices regardless of the agent's appointments with any health plan (Covered California, Agency Agreement, Scope of Work, https://www.coveredca.com/pdfs/Agency_Agreement_Monetary_Agreement.pdf). We read this as a rule of conduct rather than a description of realized behavior. An agent is still paid only by the carriers it contracts with, and we do not observe whether a given agent presents the full menu, so the provision governs what agents are obligated to do and need not describe what every agent does.

[9] As one point of comparison, in the exchange's first years Covered California paid certified enrollment counselors \$58 per application that resulted in coverage, later replacing this flat rate with individually negotiated grants (Covered California, *Marketing, Outreach and Enrollment Assistance Advisory Group*, June 12, 2014; Cal. Code Regs. tit. 10, § 6668). Broker commissions in our data average roughly \$130 per member per year and recur while the household remains enrolled (authors' calculations).

3.1 Data Sources

Enrollment data. The primary data source is individual-level administrative enrollment records from Covered California, obtained via a Freedom of Information Act (FOIA) request. The data cover the universe of exchange enrollees from 2014 to 2019 and include each enrollee's selected plan, age, income (as a fraction of the federal poverty level), county of residence, subsidy eligibility, household composition, and the type of decision assistance used (navigator, broker/agent, or none). Individual and household identifiers allow consumers to be grouped into household units and tracked across enrollment years. The enrollment data also indicate whether each household is a new enrollee or a renewal, which we use to construct our measure of market churn.

Plan characteristics. We collect plan-level data from the Covered California website, including plan premiums, metal tier, cost-sharing parameters (deductible, coinsurance, and maximum out-of-pocket limit), network type (HMO, PPO, EPO), and the insurer offering each plan. Plans are identified by a combination of insurer, metal tier, and network type within each of California's 19 rating regions. We use these data to construct each household's choice set and household-specific premiums, p_{ij}, applying the ACA age-rating curve and subsidy formula described in Section 2.

Commissions. Commission schedules are collected from Covered California's published rate schedules for each insurer and year. Commissions are set either as a flat per-member-per-month (PMPM) amount or as a percentage of the unsubsidized premium, varying by insurer but not by plan within an insurer. We convert percentage-based commissions to PMPM equivalents using each plan's average unsubsidized premium. The commission data cover all 10–12 insurers operating on the exchange in each year.

Broker density. We obtain data on the number of licensed insurance agents actively enrolling consumers in each rating region and year from a FOIA request to the California Department of Insurance (via the California Producer Licensing Authority). Broker density varies only across rating regions and years. We use it in the supplemental appendix to construct a control function for selection into assistance, as described in Section 4.2.

Rate filings and risk adjustment. For the supply-side analysis, we use plan-level rate filing data from the CMS Public Use Files (PUFs) for 2014–2020. These filings report plan-level earned premiums, incurred claims, risk adjustment transfers, and reinsurance recoveries, along with member months by plan. We use these data to estimate risk score prediction models and claims regressions that feed into the structural marginal cost decomposition (Section 5.2).

3.2 Sample Construction and Summary Statistics

Rather than merging the Covered California data with a separate survey of the uninsured, we form the universe of potential exchange consumers from the Covered California households themselves. A household observed with exchange coverage in some years is treated as uninsured, and therefore as taking the outside option, in the years of our sample in which it has no exchange plan, provided it remained eligible for the individual market in those years. Because the enrollment data do not record when a household loses eligibility, we use the Survey of Income and Program Participation (SIPP) to impute it, estimating a logit model of the probability that a household transitions into or out of individual market eligibility as a function of the demographics common to both data sets. Households predicted to have left the market are removed from the choice set in the years they are not enrolled rather than counted as uninsured. The supplemental appendix describes this construction in detail.

The enrolled sample consists of 5,396,359 insured household-year observations from 2014–2019, of which approximately 43.9% are first-time enrollees. About 64.4% of these households used some form of decision assistance when enrolling, 45.3% through a private insurance agent and 19.1% through a navigator, so agents are the more common channel and account for close to half of all enrollments. Figure 2 shows total enrollment by year and channel.

Table 1 reports summary statistics separately for assisted and unassisted households, and the two groups differ systematically. Assisted households have somewhat lower incomes (2.22 versus 2.38 times the federal poverty level) and larger households (1.46 versus 1.34 members). They skew older, with a larger share of near-elderly members aged 55 to 64 (32.5 versus 24.4 percent) and a smaller share of young adults aged 18 to 34 (26.1 versus 37.2 percent), and a somewhat lower share of male members (45.6 versus 48.3 percent). Assisted households are also more likely to be Asian (17.6 versus 12.8 percent) and slightly less likely to be Black (2.0 versus 3.0 percent), with similar Hispanic shares. Finally, assisted households are less likely to select a dominated plan, 2.2 versus 3.9 percent.

These differences are not the causal effect of assistance. Households that seek out a navigator or an agent differ from those who enroll on their own in ways correlated with plan choice, so the raw dominated-plan gap conflates the effect of assistance with the composition of who uses it. This selection on observables motivates the design-based estimator in Section 4, which reweights unassisted households to match the covariate distribution of assisted households before estimating the effect of assistance.

4 Does Decision Assistance Improve Plan Choice?

We begin by examining whether decision assistance affects plan choice and, if so, whether it improves the quality of decisions. Our estimand is the average treatment effect on the treated (ATT) for households receiving decision assistance, estimated on all enrollees so that the sample matches the structural analysis in Section 5. We embed a conditional logit discrete choice model for ACA plans into a potential outcomes framework (Rubin, 1974; Imbens and Wooldridge, 2009).

4.1 Estimation

We model plan choice with a conditional logit specification in which each enrolled household i chooses among the inside plans j available in its rating region and year, with utility

$$u_{ijt} = \alpha_{it} p_{ijt} + x'_{ij}\beta^x + \xi_j + \varepsilon_{ijt}, \tag{1}$$

where p_{ijt} is the household's subsidized premium, x_{ij} includes metal tier indicators, network type, and insurer fixed effects, ξ_j captures unobserved plan characteristics, and ε_{ijt} is an independently distributed Type I extreme value error. We allow for heterogeneous price sensitivity through $\alpha_{it} = \alpha + \phi' w_{it}$, where w_{it} includes household size, presence of children, income category, and race/ethnicity. We estimate the model conditional on enrollment, without an outside option, so the estimand is the composition of plan choice among enrollees rather than the decision of whether to enroll. Unlike the structural specification in Section 5, this model does not include commission terms, since it is estimated only on untreated households for whom commissions play no role.

Estimating this model requires constructing the full set of plan alternatives facing each household, so we draw a 20 percent random sample of households within each rating region and year and estimate on that sample. The dominated-choice analysis in Section 4.4 uses all households, since it is a binary outcome and does not require the household-by-plan choice set. Summary statistics in Table 1 describe the full population of enrollees.

We construct inverse propensity weights (IPW) to rebalance the treated and untreated groups on observable characteristics (Wooldridge, 2010). Specifically, we estimate a logit regression of assistance status on household demographics, and we assign each untreated household a weight of $\hat{p}_i/(1-\hat{p}_i)$, where $\hat{p}_i$ is the predicted probability of assistance. Treated households receive a weight of one. This reweighting ensures that the untreated group used to form counterfactual predictions is comparable to the treated group on observables.

We estimate the ATT through a four-step process. First, we estimate Equation (1) on untreated households only (those without any form of decision assistance), weighted by IPW weights. Second, we use the estimated coefficients to predict counterfactual choice probabilities for treated households, $Pr[\widehat{Y_{ij}^0|\text{Assisted}}]$. Third, we aggregate predicted probabilities across treated households to form the counterfactual enrollment count $\widehat{N_j^0}$. Fourth, we compute the ATT as the difference between observed and predicted enrollment,

$$ATT_j = \underbrace{N_j^1}_{\text{Observed}} - \underbrace{\widehat{N_j^0}}_{\text{Predicted}} . \tag{2}$$

The estimator is nonparametric in the assistance effect. Assistance enters nowhere in the fitted model, so the ATT is not a coefficient on an assistance indicator but the gap between what assisted households actually chose and what the unassisted choice model predicts for them, aggregated to the metal tier or insurer level. We obtain confidence intervals by bootstrapping the choice model and the ATT computation, resampling households with replacement within region-year cells so that cell sizes are preserved. The propensity weights are held at their full-sample values across bootstrap draws.[10]

[10]Because the propensity score is estimated from a very large number of households, its contribution to

4.2 Identifying Assumptions

The central identification challenge is that the decision to seek assistance is not random. Households who use a broker or navigator may differ systematically from those who do not, both in observable characteristics and in unobservable ways that also affect plan preferences. Our design addresses the observable component directly and assesses the unobservable component as a robustness exercise.

Identification of the ATT requires that, conditional on the household characteristics entering the propensity score and the plan characteristics entering the choice model, assisted and unassisted households would have chosen among the same plans in the same way absent assistance. A violation would require households to select into assistance on preferences that are not captured by their demographics, income, or household composition, such as an unobserved taste for comprehensive coverage that also draws a household toward a broker. If such selection were present, and if households with stronger tastes for coverage are also more likely to seek help, our estimates would overstate the effect of assistance on silver and gold enrollment. We assess the reasonableness of the conditional independence assumption in two ways. Figures 3 and 4 document common support and covariate balance after reweighting. The supplemental appendix then reports a sequence of pooled specifications in which we progressively add demographics, year fixed effects, region fixed effects, and a control function for selection into assistance built from regional variation in broker density.

We note that broker density has limited power to speak to selection within a rating region. The instrument varies only across region-year cells, so a specification that absorbs region fixed effects leaves it with essentially no variation, and within a region the number of licensed agents is close to a deterministic function of the year. When we cluster at the level at which the instrument actually varies, the partial F statistic on broker density is 3.6, against a value of roughly 7,600 when standard errors ignore the clustering entirely. We therefore report the control function and the region fixed effects as separate specifications rather than combining them, and we treat the stability of the assistance coefficients across the sequence, rather than the control function coefficient itself, as the evidence on selection.

the sampling variability of the ATT is small, so we hold the weights fixed across bootstrap draws rather than re-estimating the score within each draw.

4.3 Effects on Plan Choice

Figure 3 presents the propensity scores from the preliminary logit regression of assistance on household demographics. Common support holds across treated and untreated groups. Figure 4 shows standardized mean differences before and after reweighting. The IPW-adjusted differences are near zero for all covariates, confirming that the reweighting achieves adequate balance.

We summarize ATT estimates by insurer and metal tier in Figures 5 and 6. Figure 5 shows that assisted households are 11.9 percentage points (95% CI 11.7 to 12.0) less likely to enroll with Kaiser than the choice model predicts, and more likely to enroll with Blue Shield (5.6 percentage points), Health Net (3.1 percentage points), or a smaller insurer (2.7 percentage points). Anthem is essentially unchanged at 0.5 percentage points. Kaiser pays the lowest broker commission among the major insurers throughout our sample period, a flat \$6.25 per member per month against \$9 to \$16.50 for Anthem and percentage-based schedules for Blue Shield and, through 2017, Health Net that translate into larger per-enrollee payments (Figure 1). The ordering is consistent with commission variation shaping the plans that assisted households end up in, which we investigate more fully in Section 5.

Figure 6 shows that assisted households are 12.3 percentage points (95% CI 12.1 to 12.5) more likely to select a silver plan, an observed silver share of 69.0% against a counterfactual share of 56.7%. Most of this shift comes from a corresponding 9.5 percentage point decrease in bronze plan selection, with a smaller 2.7 percentage point reduction in gold. Platinum enrollment is unchanged. Because enhanced silver plans carry the cost-sharing reductions available to households below 250% of FPL, a shift out of bronze and into silver is the direction that decision assistance would be expected to move choices if it primarily resolves an information problem.

4.4 Effects on Dominated Choices

The plan choice results establish that assistance shifts enrollment across plans and insurers, but they do not reveal whether these shifts represent improvements. To examine choice quality, we identify specific instances in which the observed plan choice is dominated by another option in the household's choice set.

Dominance arises from the interaction of cost-sharing reductions (CSRs) with metal tier pricing. For households eligible for CSRs with incomes below 150% of FPL, enhanced silver plans offer both lower premiums and lower out-of-pocket costs than gold or platinum plans. Since provider networks do not vary across metal tiers within the same plan type, gold and platinum plans are strictly dominated for these households. Gold plans are similarly dominated for households with incomes between 150% and 200% of FPL.

We estimate the ATT of assistance on dominated choices with the same prediction-based estimator used for plan choice. We restrict the sample to households with at least one dominated plan in their choice set, fit an IPW-weighted logistic regression of the dominated-choice indicator on household demographics and year fixed effects among unassisted households, and predict the counterfactual dominated-choice rate for assisted households. As in the plan choice model, no control function enters the estimation, and assistance does not appear as a regressor.

Results are presented in Figure 7. We estimate that assisted households are 1.6 percentage points (95% CI 1.5 to 1.6) less likely to make a dominated choice than the model predicts they would have been absent assistance, an observed dominated-choice rate of 2.17% against a predicted rate of 3.73%. The effect is roughly twice as large among households using a private insurance agent (ATT of -1.8 percentage points) as among those using a navigator (ATT of -0.9 percentage points). This differential suggests that brokers, despite the potential for commission-driven steering, provide more effective guidance on average than navigators in helping households avoid dominated plans. The supplemental appendix reports the same estimates restricted to new enrollees, where the effects are somewhat larger (-2.3 percentage points overall), along with a sequence of pooled linear probability specifications.

5 Insurer Pricing with Commission-Based Steering

The design-based evidence in Section 4 establishes that broker assistance shifts plan choice and reduces dominated decisions. However, broker assistance is not free. Insurers pay commissions to brokers for each enrollment, and these commission costs are ultimately reflected in premiums paid by all enrollees, including those who do not use a broker. The net welfare effect of broker intermediation therefore depends on the tradeoff between improved plan selection for assisted households and higher premiums for the full market. To quantify this tradeoff, we develop a model of insurer premium setting in which commissions serve as the mechanism through which insurers influence the recommendations of broker-assisted consumers.

On the supply side, insurers set premiums in Bertrand-Nash competition and choose the commissions they pay brokers. On the demand side, households choose the plan that maximizes expected utility. We do not explicitly model the actions of individual insurance agents. Instead, we model how insurer commission payments affect the plan recommendations that agents make to their clients.

5.1 Household Plan Choice

Household i chooses the plan j that maximizes indirect utility

$$\begin{aligned} U_{ijt} = \alpha_{it} p_{ijt} + \beta^{\eta} \eta_{jt} a_{it}^{B} + MT_j' \left(\beta^{N} a_{it}^{N} + \beta^{B} a_{it}^{B} \right) \\ + MT_j' \Gamma w_{it} + x_{ij}' \beta^{x} + \xi_j + \varepsilon_{ijt}, \end{aligned} \tag{3}$$

where p_{ijt} is household i's subsidized premium for plan j in year t, η_{jt} is the per-member-per-month commission paid to brokers enrolling consumers in plan j, a_{it}^{B} and a_{it}^{N} are indicators for whether household i used a private broker or a navigator, MT_j is a vector of metal tier indicators, w_{it} is a vector of household demographics, x_{ij} is a vector of plan and household-plan interaction characteristics, ξ_j captures unobserved plan characteristics, and ε_{ijt} is an error term.[11]

Each of the two assistance channels enters utility separately via channel-by-metal interactions $MT_j'(\beta^{N} a_{it}^{N} + \beta^{B} a_{it}^{B})$, which allow navigators and brokers to shift metal tier preferences by different amounts. The commission term $\beta^{\eta} \eta_{jt} a_{it}^{B}$ enters utility only for broker-assisted households, since navigators receive no plan-specific compensation, and a positive β^{η} indicates that brokers steer consumers toward plans with higher commissions. For unassisted households, all assistance-related terms are zero, and choices are driven entirely by premiums,

[11] While inertia is known to play an important role in health insurance plan choice (Handel, 2013; Polyakova, 2016), it is less of a concern for our analysis. Approximately 43.9% of insured household-year observations in our sample are first-time enrollees, consistent with the high churn rates characteristic of ACA individual market coverage. The supplemental appendix reports the design-based assistance effects re-estimated on new enrollees, where inertia plays no role, and finds the same pattern with somewhat larger magnitudes.

plan characteristics, and unobserved preferences.

Attaching the commission term to every plan in a broker-assisted household's choice set treats the household as selecting a broker and a plan together rather than in a fixed order. A household that is already aware of a few plans and consults an agent as it nears a decision, and a household that chooses an agent and a plan at the same time, both reduce to a broker-channel household whose choice over the full menu is tilted by the commissions attached to those plans. The alternative in which a household first commits to a particular agent and is then confined to the carriers that agent represents would require the identity of each household's agent and that agent's carrier contracts, neither of which we observe. Because commissions enter the choice over the entire menu, β^{η} measures the average pull of commissions on broker-assisted choices, and to the extent that some agents cannot enroll clients in every plan, treating the full menu as commission-relevant attenuates β^{η} toward zero rather than inflating it.

The demographic-by-metal interactions $MT_j'\Gamma w_{it}$ let the valuation of coverage generosity vary with household age composition and gender composition. These are not a channel distortion but a feature of preferences, and they matter for the supply side. Without them the model sorts households across metal tiers almost entirely through the price coefficient, predicted enrollee composition varies too little across plans within a market, and the risk score equation in Section 5.2 cannot separate the demographic composition of a plan's enrollees from its generosity.

Insurers set commissions either as a flat per-member-per-month amount or as a percentage of the unsubsidized premium. For percentage-based commissions, we convert to per-member-per-month equivalents using the plan's average unsubsidized premium. In practice, commission schedules exhibit remarkably little temporal variation. Of the ten insurers operating on the exchange during our sample period, seven maintained identical commission rates across all years. Only the three largest insurers (Anthem, Blue Shield, and Health Net) adjusted their rates, and these adjustments followed a monotone downward trend rather than the year-to-year reoptimization that a commission first-order condition would imply. We return to the insurer's commission choice in the counterfactuals of Section 6, where we let commissions re-optimize according to the first-order condition in Equation (13).[12]

[12]Health Net's switch from percentage-based to flat commissions in 2018 followed its acquisition by Centene

The household's subsidized premium is

$$p_{ijt} = \max\left\{ \underbrace{\sigma_{it} p_{jmt}}_{\text{full premium}} - \underbrace{\max\{\sigma_{it} p_{bmt} - \zeta_{it}, 0\}}_{\text{premium subsidy}}, 0 \right\}, \tag{4}$$

where σ_{it} is the household's age-based rating factor, p_{jmt} is the base premium for plan j in market m at time t, p_{bmt} is the base premium of the benchmark plan (the second-cheapest silver plan), and ζ_{it} is the household's income contribution cap. Premium subsidies are available to households with incomes between 100% and 400% of the federal poverty level. The income contribution cap ranged from 2% of annual income at 100% FPL to 9.5% at 400% FPL in 2014, with slight annual increases. Subsidies can be applied toward any non-catastrophic plan. For some low-income consumers, the subsidy may exceed certain bronze plan premiums, in which case the premium is set to zero.

We allow for heterogeneous price sensitivity by parameterizing

$$\alpha_{it} = \alpha + \phi' w_{it} + \alpha^N a_{it}^N + \alpha^B a_{it}^B,$$

where w_{it} includes household size, age composition, gender, income category, and race and ethnicity. In estimation this amounts to including interactions between p_{ijt} and each element of w_{it}, together with interactions between p_{ijt} and the two channel indicators, so that assisted households are allowed a different price response than unassisted households rather than being forced to share one. The vector x_{ij} includes metal tier indicators, network type, an indicator for health savings account eligibility, and fixed effects for the four largest insurers.[13] The utility of the outside option is $U_{i0t} = \alpha_{it}\rho_{it} + \varepsilon_{i0t}$, where ρ_{it} is the household's penalty for not having insurance.

Identification of the assistance effects. The assistance parameters are identified from variation across the plans in a household's choice set rather than from variation in who obtains assistance. Household-level characteristics, including the assistance indicators them-

and represents an organizational change rather than marginal optimization.

[13]The seven regional insurers remain separate plans with their own premiums, commissions, and costs, but carry no brand fixed effect. Adding one fixed effect per regional insurer drives the nesting parameter above one, outside the range consistent with random utility maximization.

selves, are differenced out of the conditional choice probabilities, so an assistance term can affect choices only through its interaction with a plan characteristic. The metal and commission terms are therefore identified by how the plan rankings of assisted and unassisted households differ within a market and year, holding the choice set fixed. Commission steering is separately identified from general guidance because commissions vary across insurers within a metal tier and, for the three largest insurers, across years within an insurer, while the metal terms are common across insurers within a tier.

This does not resolve the possibility that households select into assistance on unobserved preferences. We do not instrument for assistance in the structural model. A control function residual is constant within a household, so it can enter the conditional logit only through an interaction with a plan characteristic, and the interaction we would use is collinear with the commission level term it is meant to correct. The design-based evidence in the supplemental appendix bears directly on how much this matters. The assistance coefficient in the pooled dominated-choice specifications is stable as controls accumulate, and the control function specification recovers the same total effect while splitting it into two large offsetting pieces, which is what a weak first stage produces rather than evidence of selection. The commission coefficient carries a stronger identification argument still. Households never observe commissions, so there is no direct channel from household tastes to commission levels, and with insurer fixed effects absorbing brand preferences, the coefficient is identified from commission variation within insurers over time and across plans within a brand. Selection into broker use cannot manufacture that correlation. We therefore interpret the assistance parameters as causal effects of assistance on plan choice. They rest on the same conditional independence assumption the design-based analysis defends, supported by the covariate balance after reweighting, the stability of the pooled coefficients, and the new-enrollee replication in the supplemental appendix.

Choice probabilities. We assume that the error terms ε_{ijt} follow a generalized extreme value distribution, yielding a nested logit model with exchange plans in one nest and the outside option in a second nest. This two-nest structure captures the observed substitution pattern between silver plans and the outside option that arises from the ACA's linkage of cost-sharing reductions to silver plan enrollment. The household choice probabilities are

$$q_{ijt} = \frac{e^{V_{ijt}/\lambda}\left(\sum_{k \in J_{mt}} e^{V_{ikt}/\lambda}\right)^{\lambda-1}}{e^{V_{i0t}} + \left(\sum_{k \in J_{mt}} e^{V_{ikt}/\lambda}\right)^{\lambda}}, \tag{5}$$

where V_{ijt} is the deterministic component of utility from Equation (3), $V_{i0t} = \alpha_{it}\rho_{it}$ is the deterministic utility of the outside option, J_{mt} is the set of exchange plans in market m at time t, and $\lambda \in (0, 1]$ is the nesting parameter. The choice probabilities converge to standard logit when $\lambda \to 1$.

The sensitivity of demand to a base premium change involves the chain rule through the subsidy formula. For a non-benchmark plan, an increase in its base premium raises only that plan's consumer-facing premium. For the benchmark plan, an increase raises the subsidy available to all plans, leaving the benchmark plan's consumer-facing premium unchanged for subsidized households while reducing the effective premium of all competing plans. Specifically,

$$\frac{\partial p_{ilt}}{\partial p_{jmt}} = \begin{cases} 0 & l = j,\ j = b \\ \sigma_{it} & l = j,\ j \neq b \\ -\sigma_{it} & l \neq j,\ j = b \\ 0 & l \neq j,\ j \neq b \end{cases}, \tag{6}$$

where b denotes the benchmark plan. This subsidy linkage creates substantial computational challenges but is essential to model given the central role of premium subsidies in ACA markets.

Identification of the premium parameter. Several sources of exogenous variation identify the premium coefficient α_{it}. First, the phasing-in of the individual mandate penalty between 2014 and 2016, and the elimination of the penalty in 2019, generate variation in the relative price of the outside option. Second, kinks in the subsidy formula (4) generate variation in relative premiums across plans. Because the subsidy is tied to the benchmark plan premium, some bronze plans may be free to low-income consumers when the subsidy exceeds the full bronze premium. The set of free plans varies by market, year, and household characteristics. Finally, we include insurer fixed effects to absorb unobservable plan characteristics such as provider networks and formularies that may correlate with premiums, following Ho and Pakes (2014) and Tebaldi (2025).

5.2 Insurer Pricing

Taking commission schedules as given, each insurer f chooses base premiums p_{jmt} for each of its plans to maximize expected variable profit

$$\pi_{ft} = R_{ft} - (1 - \iota_{ft})C_{ft} - \mathcal{C}_{ft} + RA_{ft} - V_{ft}, \tag{7}$$

where R_{ft} is total premium revenue, C_{ft} is total claims, ι_{ft} is the reinsurance rate (the expected share of claims covered by the ACA transitional reinsurance program), $\mathcal{C}_{ft}$ is total commissions paid, RA_{ft} is the net risk adjustment transfer, and V_{ft} is variable administrative cost excluding commissions. Total commissions paid by the firm are

$$\mathcal{C}_{ft} = \sum_{m \in M} \sum_{j \in J_{fmt}} \sum_{i \in I} a_{it}^{B} \, q_{ijt} \, \eta_{jt},$$

where a_{it}^{B} indicates broker-assisted enrollment and q_{ijt} are the choice probabilities from Equation (5).

Risk adjustment. The ACA risk adjustment program makes budget-neutral transfers across plans based on the relative health risk of their enrollees (Pope et al., 2014). Plans that attract higher-risk enrollees receive transfers funded by plans with lower-risk enrollees. The per-member-per-month transfer for plan j is

$$ra_{jmt} = \left(\frac{r_{jmt}}{\sum_l r_{lmt} \tilde{q}_{lmt}} - \frac{h_j}{\sum_l h_l \tilde{q}_{lmt}} \right) \bar{p}_t, \tag{8}$$

where r_{jmt} is the plan risk score, h_j is an exogenous utilization factor that accounts for the plan's actuarial value and associated moral hazard, $\tilde{q}_{lmt}$ denotes enrollment-weighted plan shares, and $\bar{p}_t$ is the market average premium. The first term in parentheses captures the plan's share of total risk (reflecting adverse selection, moral hazard, and plan generosity), while the second captures the plan's expected utilization share (reflecting only moral hazard and generosity). A positive difference indicates that the plan attracts higher-risk enrollees than its generosity alone would predict, generating an inward risk adjustment transfer. Budget neutrality is maintained by construction.

We do not directly observe plan risk scores in the rate filing data. Instead, we estimate a risk score prediction model as a function of the plan's actuarial value and the demographic composition of its enrollees,

$$\ln r_{jmt} = \gamma^{AV} AV_j + \sum_{d \in D} \gamma^d s_{djmt} + \sum_f \gamma^f \mathbb{1}[f(j) = f] + \epsilon^r_{jmt}, \tag{9}$$

where AV_j is the plan's actuarial value, s_{djmt} is the share of plan j's enrollment in demographic group d, and the final term is a set of insurer fixed effects. Actuarial value replaces the metal tier indicators used by Saltzman (2019) because it subsumes them while also capturing variation within the silver tier across the three cost-sharing reduction levels. The demographic shares are the age and gender shares of enrollment. To avoid endogeneity from selection on unobservables, we compute predicted demographic shares by aggregating the estimated household choice probabilities from Equation (5) rather than using observed enrollment shares, following the approach widely used in the hospital choice literature to construct market concentration measures (Kessler and McClellan, 2000).

The insurer fixed effects in Equation (9) are a deliberate departure from the specification in Saltzman (2019), and they are data-driven. Without them, the low-cost carriers in the market sit roughly 0.4 to 0.7 log points below the risk score that actuarial value and demographics alone predict for them. The model then overstates their enrollees' risk, pays them risk adjustment transfers in excess of their claims, and drives their implied marginal costs negative, even though these carriers pay into the risk adjustment pool in the raw filings. Adding insurer fixed effects corrects the mislabeling. Because risk adjustment is budget neutral within a market, it does not reduce the total number of plans with negative net-of-transfer marginal cost, and we do not treat that count as a diagnostic. Roughly nine percent of platinum plans in the raw rate filings genuinely receive more in risk adjustment and reinsurance than they pay in claims, so an efficient insurer covering a high-risk pool can book a surplus net of transfers.

We predict plan average claims as a function of the predicted risk score,

$$\ln c_{jmt} = \mu^r \ln \hat{r}_{jmt} + x_j' \mu^x + \mu^l l_t + \epsilon^c_{jmt}, \tag{10}$$

where $\hat{r}_{jmt}$ is the fitted risk score from Equation (9), x_j is a network type indicator, and l_t is a linear trend. Using the predicted rather than the observed risk score addresses the endogeneity of plan enrollment composition. Actuarial value and the insurer fixed effects enter the risk score equation only. Including actuarial value in both equations makes it collinear with the risk score and collapses the estimated pass-through of risk into claims from roughly one to near zero, and insurer identity enters both equations through the same pass-through, which leaves the joint moment condition rank deficient.

Pricing first-order conditions. Differentiating Equation (7) with respect to the base premium p_{jmt} yields the first-order condition

$$\frac{\partial R_{ft}}{\partial p_{jmt}} + \frac{\partial RA_{ft}}{\partial p_{jmt}} = (1 - \iota_{ft}) \frac{\partial C_{ft}}{\partial p_{jmt}} + \frac{\partial \mathcal{C}_{ft}}{\partial p_{jmt}} + \frac{\partial V_{ft}}{\partial p_{jmt}} \tag{11}$$

for each plan j offered by firm f. Collecting the first-order conditions across all plans into matrix form, the equilibrium markup for each plan is

$$p - mc = \Omega^{-1} \left(s + \Omega^B \eta \right), \tag{12}$$

where p is the vector of base premiums, mc is the vector of marginal costs (net of risk adjustment and reinsurance), s is the vector of market shares, and η is the vector of commission rates. The ownership matrix Ω has elements

$$\Omega_{jl} = -\mathbb{1}\left[f(j) = f(l) \right] \frac{\partial s_l}{\partial p_j},$$

and Ω^B is the analogous matrix constructed from broker-assisted demand derivatives only. The term $\Omega^B \eta$ captures how commission costs distort pricing relative to a standard Bertrand-Nash equilibrium without intermediaries. Plans with higher commissions and more price-sensitive broker-assisted enrollees face larger upward pressure on premiums.

5.3 Estimation

We estimate the model sequentially in three stages. First, we estimate the demand parameters $\boldsymbol{\beta}$ and the nesting parameter λ by maximum likelihood using the nested logit choice probabilities in Equation (5). We pool households across markets and years, estimating a single set of coefficients, and weight each household by its size (number of members). As in Section 4.1, demand is estimated on a 20 percent random sample of households drawn within each rating region and year. This weighting ensures that predicted market shares correspond to member-level enrollment, which is the relevant unit for insurer revenue, risk pool composition, and the supply-side first-order conditions. Second, given the estimated demand parameters, we compute market shares and the elasticity matrices Ω and Ω^B for each market-year cell and recover plan-level marginal costs by inverting the pricing first-order condition in Equation (12). Third, we estimate the cost parameters by two-step generalized method of moments, following Saltzman (2019). The moment conditions stack three blocks, namely the risk score residuals from Equation (9), the claims residuals from Equation (10), and the pricing first-order condition residuals from Equation (11) evaluated at observed premiums. The second-step weight matrix is the inverse of the estimated moment covariance rather than a block-diagonal scalar, which matters for precision as much as for point estimates.[14] We drop a plan-cell from the first-order condition block when its predicted within-cell share falls below 0.002, since the markup inversion is ill-conditioned as the share approaches zero.

We report analytical standard errors. For the demand parameters we use a sandwich estimator clustered at the household level, computed from the score and Hessian of the pooled likelihood. For the cost parameters we use the corresponding generalized method of moments sandwich, which accounts for the first-stage demand estimates entering through the predicted demographic shares and the first-order condition residuals. Standard errors for the counterfactual welfare quantities in Section 6 are obtained by a parametric bootstrap over the demand parameters, re-scoring welfare at the solved equilibrium premiums held fixed rather than re-solving the equilibrium on each draw.[15]

[14]Using a scalar weight understates the risk score standard errors by more than an order of magnitude in our data.

[15]Re-solving the pricing equilibrium on every bootstrap draw is computationally prohibitive. Holding premiums at their solved values is a small approximation, since the equilibrium premiums move by only a few dollars per plan across draws, leaving the welfare standard errors nearly unchanged.

5.4 Results

Table 2 reports the demand estimates, obtained from the 20 percent sample of 1.54 million households. The nesting parameter is 0.674 (95% CI 0.669 to 0.678), comfortably inside the unit interval required by random utility maximization and well away from the standard logit boundary, so households substitute more readily among exchange plans than between the exchange and the outside option. The base premium coefficient is -0.68 per hundred dollars of monthly premium. The channel premium interactions offset roughly two-thirds of that sensitivity for assisted households (0.46 for navigators and 0.49 for brokers), so households choosing with assistance respond far less to premium differences than households choosing on their own.

The steering terms line up with the design-based evidence. The commission coefficient is positive and precisely estimated (0.021 per dollar of monthly commission, 95% CI 0.020 to 0.021), so broker-assisted households are more likely to enroll in plans paying higher commissions, holding premiums and plan characteristics fixed. Evaluated at the mean price coefficient among broker-assisted households, a \$1 increase in monthly commission shifts their enrollment as much as a \$3.29 reduction in net monthly premium. Because mean commissions (\$11 per member per month) are an order of magnitude smaller than mean net premiums (\$91), the same comparison in percentage terms puts the commission elasticity of broker-assisted demand at roughly two-fifths of its premium elasticity. The channel-by-metal interactions raise the attractiveness of the silver and bronze tiers relative to gold and platinum for both channels, and the demographic-by-metal interactions generate the sorting of younger and male-skewed households across tiers that identifies the demographic terms in the risk score equation.

Table 3 reports the cost estimates, which follow Saltzman (2019) where the specifications overlap. Actuarial value is the dominant term in the risk score equation with a coefficient of 3.83 (95% CI 3.23 to 4.43), the predicted age shares enter negatively so that plans enrolling younger members carry lower risk scores, and the male share enters at -0.51, though imprecisely estimated. The insurer effects place the low-cost carriers well below the risk score that actuarial value and demographics alone would assign them, the mislabeling correction discussed in Section 5.2. In the claims equation, the pass-through of the risk score into average claims is 1.19 (95% CI 1.02 to 1.37), close to and slightly above one.

Table 4 reports the recovered markups by metal tier. Markups rise with plan generosity, from roughly $57 per member per month at the median bronze plan to $139 at the median platinum plan, with Lerner indices from 0.20 for bronze to 0.28 for platinum. Figure 8 plots the marginal cost implied by the pricing first-order conditions against the structural prediction from the risk score and claims equations. The correlation between the two is 0.50 among the 2,344 plan-cells above the share floor. Of those, 72 carry negative net-of-transfer marginal cost, all in Kaiser's gold and platinum plans, consistent with the raw-filings pattern discussed in Section 5.2. The remaining gap between the two cost measures is absorbed by the plan-level cost residual in the counterfactuals, so the equilibrium comparisons in Section 6 do not depend on the fit of the cost model.

6 Welfare Effects of Broker Intermediation

The design-based evidence in Section 4 shows that broker assistance improves plan choice, while the structural estimates in Section 5 quantify how commissions drive broker recommendations and affect insurer pricing. In this section, we use the estimated model to evaluate the net welfare effects of broker intermediation through a series of counterfactual simulations. Each counterfactual solves for a new Nash equilibrium in which insurers re-optimize premiums in response to a change in the commission environment, enrollment reshuffles across plans, risk pools adjust, and risk adjustment transfers rebalance accordingly.

6.1 Counterfactual Design

We consider a set of counterfactual scenarios that isolate different aspects of the welfare tradeoff. The scenarios fall into two groups. In the first, we impose a counterfactual commission schedule and let insurers re-optimize premiums alone. In the second, insurers re-optimize premiums and commissions jointly, so that the commission schedule itself responds to the policy change.

Zero commissions. In the first scenario, we set all commissions to zero and solve for the new equilibrium premium vector. This scenario has two effects. First, insurers' marginal costs fall by the amount of eliminated commission payments, generating downward pressure on premiums. Second, the steering channel in the demand model is shut off ($\eta_j = 0$ for all j), so broker-assisted households no longer receive commission-driven plan recommendations. Households who previously used a broker may still receive assistance (from a navigator, for instance) or may revert to unassisted choice. The welfare implications depend critically on what replaces broker assistance.

To capture this, we parameterize the substitution from broker assistance to navigator assistance using a substitution rate $\tau \in [0, 1]$. At $\tau = 0$, all formerly broker-assisted households become unassisted. At $\tau = 1$, all switch to navigator assistance, retaining the benefits of guidance without the commission-driven steering. For intermediate values, a fraction τ of broker-assisted households transition to navigators, with the remainder becoming unassisted. We allocate the τ fraction based on each household's estimated propensity to use a navigator, computed from a logistic regression of navigator use among assisted households. This propensity-based allocation reflects the idea that households most similar to current navigator users are most likely to substitute toward navigator assistance.

We solve the zero-commission equilibrium at each point on a grid of τ values ranging from 0 to 1 in increments of 0.25. At each τ, insurers re-optimize premiums given the modified demand system. Comparing welfare across the τ gradient decomposes the total effect of removing commissions into two components. The gap between welfare at $\tau = 1$ and the observed equilibrium captures the pure premium savings from eliminating commission costs. The gap between welfare at $\tau = 0$ and $\tau = 1$ captures the value of assistance itself, holding premiums at their zero-commission equilibrium levels.

Uniform and scaled commissions. We next replace the observed schedule with a uniform commission equal to the mean observed commission among plans with positive commissions, which removes the cross-insurer variation that drives differential steering while preserving the overall level of compensation. A related set of scenarios scales every commission to a common fraction of its observed value, at one-quarter, one-half, and three-quarters, tracing how the premium and welfare effects grow with the level of compensation. Comparing these equilibria to the observed equilibrium separates the welfare cost of differential steering from the premium-level effects of commissions.

Aligned commissions. A further scenario reallocates commissions across plans in proportion to each plan's non-commission consumer value, the mean over the cell's households of their indirect utility from the plan with the commission term removed, rescaled to hold the total commission budget fixed at its observed level. This keeps compensation constant but ties it to consumer fit rather than to insurer margin, so that brokers steer toward the plans households value most. It isolates the alignment of commissions from their level.

Endogenous commissions. The remaining scenarios let insurers choose commissions as well as premiums. Each insurer f scales its observed commission schedule by a factor k_f, so that the commission on plan j becomes $\eta_{jt} = k_f w_{jt}$, where w_{jt} is the plan's commission basis, a flat per-member amount for flat schedules and the observed rate times the plan's premium for percentage schedules. Differentiating the variable profit in Equation (7) with respect to k_f yields the commission first-order condition

$$\sum_{j \in J_{ft}} (p_{jt} - mc_{jt} - \eta_{jt}) \frac{\partial q_{jt}^B}{\partial k_f} + \frac{\partial RA_{ft}}{\partial k_f} = (1 + \mu_f) \sum_{j \in J_{ft}} w_{jt} \, q_{jt}^B, \tag{13}$$

where q_{jt}^B is broker-assisted enrollment in plan j and J_{ft} is the set of plans offered by insurer f. The left-hand side is the marginal benefit of a higher commission, the additional variable profit from the broker-assisted enrollment the commission attracts, which operates through the steering coefficient β^η, together with the induced change in risk adjustment transfers. The right-hand side is the marginal cost, the additional commission outlay, scaled by a per-insurer wedge μ_f that represents the shadow value of a commission dollar. We calibrate μ_f so that the observed schedule ($k_f = 1$) satisfies Equation (13), which makes the observed premiums and commissions a fixed point of the joint system and reconciles the endogenous treatment here with the predetermined treatment in estimation, since the observed commissions are by construction optimal given μ_f. We hold μ_f fixed across scenarios, in the same spirit as the plan-level cost residual introduced in Section 6.2, and solve Equation (13) jointly with the pricing first-order conditions in Equation (11), so that premiums and the commission scales $\{k_f\}$ are determined together.

Navigator expansion. We expand navigator access by converting a fraction τ of broker-assisted households to navigators while the remaining brokers continue to operate and be paid, with insurers re-optimizing commissions in response. Unlike the zero-commission scenario, the broker market is not eliminated, so this measures the effect of widening public assistance alongside a functioning private channel.

Flat-fee mandate. We require the percentage-of-premium insurers to compensate brokers with a flat per-member amount instead, with the level chosen endogenously. This removes the interaction between commissions and premiums that a percentage schedule creates, under which raising a premium mechanically raises the commission the plan pays.

Navigator defunding. We convert a fraction of navigator-assisted households to the broker channel, the direction of the federal navigator funding cuts between 2017 and 2019, again with commissions re-optimized. This traces the welfare consequences of contracting public assistance and shifting households toward commissioned agents.

Observed commissions. As a benchmark, we verify that the model reproduces observed premiums when commissions are set to their actual values, which serves as a calibration check on the equilibrium solver.

6.2 Equilibrium Computation

Each counterfactual requires solving for a Nash equilibrium in which all insurers simultaneously set premiums to satisfy the first-order conditions in Equation (11), given the modified commission schedule and demand system. Because risk adjustment transfers and marginal costs depend on enrollment composition, which in turn depends on premiums, we solve the full system jointly rather than iterating between pricing and cost updates. Specifically, at each candidate price vector we recompute enrollment shares, demographic composition, predicted risk scores, claims, risk adjustment transfers, and marginal costs, so the first-order conditions embed the complete price-cost feedback loop.

We solve this system of nonlinear equations by Broyden's method with a Levenberg-Marquardt hook step, starting from observed premiums. Markets containing near-substitute plans produce Jacobians with condition numbers on the order of 10^6, and the hook step is what keeps those cells stable.

Before solving any counterfactual, we calibrate a plan-level cost residual ω_{jmt} so that the first-order conditions hold exactly at observed premiums, in the spirit of the marginal cost recovery in Berry et al. (1995) and Nevo (2000). The residual is the part of marginal cost that the cost model in Section 5.2 does not explain, and we hold it fixed across scenarios. Marginal cost in each counterfactual is then the model-predicted cost at the candidate premium vector plus this fixed residual. This makes the observed-commission scenario reproduce observed premiums by construction, so differences across scenarios reflect the change in the commission environment rather than the fit of the cost model.

Premium subsidies are also endogenous to the equilibrium. When the benchmark plan's premium changes, the subsidy available to all households adjusts according to the ACA formula in Equation (4). We update subsidies at each pricing iteration, so the equilibrium premiums, subsidies, shares, and risk adjustment transfers are jointly consistent.

6.3 Welfare Measurement

Welfare measurement in a steering model requires a position on whose preferences count. The estimated utility function describes the choices households actually make, including the part of those choices that reflects broker steering. Evaluating welfare with that utility function treats a commission-driven recommendation as a genuine benefit, which is the wrong benchmark for the policy question. We therefore report three measures.

The first is the standard Small-Rosen consumer surplus from the nested logit model (Small and Rosen, 1981),

$$CS_i = \frac{1}{\bar{\alpha}} \ln \left(e^{V_{i0}} + \left(\sum_{j \in J} e^{V_{ij}/\lambda} \right)^{\lambda} \right), \tag{14}$$

where V_{i0} is the deterministic utility of the outside option and $\bar{\alpha}$ is the marginal utility of income used to convert utility into dollars, held common across households and fixed across scenarios. We set it to the household-size-weighted mean of the base premium coefficient, with the channel price interactions removed. Using a common denominator keeps the comparison across counterfactual scenarios on a consistent basis, since the premium coefficient otherwise varies by assistance level and would re-price the baseline inclusive value in each scenario. We report consumer surplus per member per year, on the same basis as the objective measure, and treat it as the benchmark for how households themselves would evaluate a change.

The second and third measures instead value the plan the household actually ends up choosing under a normative decision rule, taking the household's choice probabilities from the full estimated model but replacing the valuation of the chosen plan. For household i we compute $\sum_j q_{ijt} V_{ijt}^N$, where q_{ijt} are the estimated choice probabilities and V_{ijt}^N is a normative valuation of plan j.

Under the first normative rule, we impose the navigator decision rule on every household, and we denote the resulting valuation V^{nav}. The navigator price slope and metal valuations are folded into the base preferences, and the commission and broker-specific terms are set to zero. This asks what households would be worth if every household were advised as a navigator advises, while still choosing plans as they in fact do. The rule is a fixed normative benchmark rather than a claim that a navigator is present, so it remains well defined in scenarios that reduce or remove navigator assistance. Under the second, we value each plan in dollars as the negative of the household's annual premium plus the certainty equivalent of its out-of-pocket exposure,

$$V_{ijt}^{obj} = -\left(\text{premium}_{ijt} + \mathbb{E}[OOP_{ijt}] + \tfrac{\rho}{2}\text{Var}[OOP_{ijt}]\right), \tag{15}$$

where the out-of-pocket distribution is generated by applying the plan's actual cost-sharing schedule, namely the deductible, coinsurance rate, and out-of-pocket maximum, to a lognormal distribution of annual spending. This is feasible because Covered California uses standardized benefit designs, so cost-sharing is determined by metal and cost-sharing-reduction tier rather than approximated by one minus the actuarial value. Expected spending varies with the household's age composition and income bracket, using survey-weighted cell means of total annual expenditure computed from the 2018 Medical Expenditure Panel Survey full-year consolidated file, and reflects who the household is rather than which plan it holds, so the measure carries no moral hazard response. Two parameters of this measure are calibrated from the literature rather than estimated. We set the coefficient of variation of annual individual spending to 2.5, a lower bound from published tabulations of spending concentration, and the coefficient of absolute risk aversion to 2.3×10^{-4} per dollar, the mean estimate in Handel (2013). This valuation applies to the insured plans, for which cost-sharing is defined by the standardized benefit design.

We value the uninsured option differently, since an uninsured household faces no plan cost-sharing. We do not value it at the household's full medical spending. That would overstate the burden, because the uninsured do not in fact pay the catastrophic bills that instead become uncompensated care, and it would place the uninsured option far outside the range of any insured plan. We instead value the uninsured option at the out-of-pocket amount uninsured households actually pay, taken by age and income from the 2018 Medical Expenditure Panel Survey, plus three costs of lacking coverage that the realized bill does not capture. The first is the value of the risk protection insurance provides beyond the bill itself, which Finkelstein et al. (2019) recover from the Oregon Health Insurance Experiment. The second is a mortality cost, formed as the household's age-specific baseline mortality times the proportional reduction in mortality associated with coverage, which Miller et al. (2021) and Goldin et al. (2021) estimate, valued at the federal value of a statistical life.[16] The third is a financial-distress cost applied to the share of uninsured households whose spending exceeds forty percent of income, the standard catastrophic-expenditure threshold. The mortality term is both the largest of the three and the least certain, since the underlying mortality effect of coverage is imprecisely estimated. We therefore report the objective welfare effects as a band across low, central, and high values of the cost of being uninsured rather than as a single number, and we hold this calibration fixed across scenarios so that it does not interact with the equilibrium.

We aggregate across households, weighting by household size. The counterfactual reports the quantities the estimated parameters drive, namely the change in the share of households that lose coverage and the welfare of those who remain insured, and the cost of being uninsured is applied to those quantities afterward. This keeps two kinds of uncertainty separate.

[16] We apply the U.S. Department of Health and Human Services guideline value, roughly \$13 million in 2023 dollars, uniformly across income. Applying a common value rather than scaling it by income avoids assigning lower-income households a smaller value of mortality-risk reduction simply because they would pay less for it.

The sampling uncertainty in the estimated demand parameters, which we obtain from a parametric bootstrap that holds the solved equilibrium premiums fixed, enters through the coverage and composition quantities. The cost parameters affect welfare only through premiums, which are fixed in the bootstrap, so they do not enter these standard errors. The calibration uncertainty in the cost of being uninsured enters through the low-to-high band. Because the uninsured cost is a fixed calibration rather than an estimated quantity, the bootstrap is run once and the band is applied to the same bootstrapped quantities.

6.4 Results

Table 5 reports each scenario's effect on premiums, on the share of households that lose coverage, and on welfare relative to the observed equilibrium, and Table 6 reports the objective welfare effect across the low, central, and high valuations of being uninsured. Table 7 reports the coverage effect and the objective welfare effect with bootstrap standard errors from a parametric bootstrap over the estimated demand parameters. All dollar figures are per member per year.

We begin with the effect on coverage. When commissions are banned and the formerly broker-assisted households receive no substitute ($\tau = 0$), the uninsured share rises by 7.7 percentage points. This shift does not depend on how we value being uninsured, and it points in one direction, since a household that loses coverage is worse off on access, health, and financial protection. When the withdrawn brokers are replaced by navigators, nearly all of the coverage returns, and the uninsured share sits only 0.9 percentage points above the observed level. The coverage loss therefore comes from the withdrawal of broker assistance itself rather than from the removal of commissions. Consumer surplus moves in the same direction, falling sharply when assistance is withdrawn and recovering most of the way when brokers are replaced by navigators, consistent with households valuing the guidance in their own choices. The navigator rule diverges from these two measures. It falls most when assistance is removed, since households left unassisted make the choices a well-informed advisor would steer them away from, but it rises when brokers are replaced by navigators, since that substitution removes the commission steering while preserving the guidance.

The welfare value of that coverage loss depends on how heavily we weight the cost of being uninsured. At the lightest weighting, we value the uninsured option only at an estimated out-of-pocket cost from the MEPS, which falls well below the full cost of care since a large share of medical care for this population is ultimately uncompensated. On this basis, removing assistance is a welfare gain of roughly \$350 per member per year under the objective measure, which values households by premium, expected out-of-pocket cost, and risk rather than by revealed preference, because the premium savings to households that remain insured exceed this modest out-of-pocket cost. Adding the value of risk protection and the mortality cost reverses this result. At the central valuation the objective effect is a loss of roughly \$390 per member per year, and it ranges from a small gain of about \$120 when these consequences are valued at the low end to a loss of about \$680 at the high end. The width of this band, and the sign of the effect within it, reflect the mortality valuation almost entirely, so that removing assistance is a welfare loss at the central and high valuations and a slight gain only when the health consequences of losing coverage are valued at the low end.

This welfare loss operates through coverage rather than through which plan households choose. To isolate the two margins, we hold premiums fixed and remove all decision assistance, both navigator and broker, and split the resulting welfare change into a coverage part, households leaving the market, and a plan-choice part, reshuffling among those who remain insured. This exercise removes more than the commission ban above, all assistance rather than the broker channel alone, and holds premiums fixed rather than letting insurers re-price, so its total of about \$647 per member per year is larger, but it cleanly attributes the effect to a margin. The coverage part accounts for essentially the entire change, a loss of about \$725, while the plan-choice part slightly offsets it with a gain of about \$77, because on the objective metric assistance steers households toward silver plans whose higher premium roughly matches their out-of-pocket and risk benefit. Table 9 reports the decomposition and traces the coverage effect across the nesting parameter and the value of the outside option, where it ranges from roughly 6 to 17 percentage points, so its direction and order of magnitude are robust while its precise level depends on those modeling choices.

Reforms that change the commission schedule without withdrawing assistance have small effects throughout. A uniform commission, commissions aligned with consumer value, and a flat-fee mandate for the percentage-based insurers each move the objective measure by less than \$20 per member per year and leave the coverage share essentially unchanged, and the scaled-commission scenarios show the effect growing smoothly with the level of compensation. The welfare cost of the differential component of commissions, apart from the broker channel itself, is small.

We turn next to the scenarios in which insurers re-optimize commissions, which speak most directly to policy. Expanding navigator access alongside a functioning broker market raises the uninsured share slightly and lowers the objective measure by roughly $53 per member per year at the central valuation, consistent with navigators being somewhat less effective than brokers at moving households into coverage and keeping them there. Defunding navigators and shifting those households to brokers, the direction of the federal funding cuts between 2017 and 2019, lowers the uninsured share by 1.1 percentage points and raises the objective measure by about $60 per member per year. The gain from defunding grows with the assumed cost of being uninsured, since its main effect is to reduce the uninsured share, and it becomes a small loss only at the low end of the band. The navigator rule moves the other way, since the households shifted to brokers now make more of the steered choices that rule penalizes, so whether defunding improves welfare depends on which measure one adopts.

Table 8 reports the producer-surplus and government-cost sides of the same scenarios. Because the subsidy is paid only on enrolled households, government spending tracks the coverage effect, falling when assistance is withdrawn and rising when households are shifted toward brokers. The producer-surplus column shows the corresponding change in insurer margins net of the commissions that fund the broker channel.

Taken together, the counterfactuals point to one conclusion. The primary effect of decision assistance is to keep households insured. Removing it raises the uninsured share substantially, and whether this amounts to a large welfare loss or a small one depends on the cost assigned to being uninsured, which we report as a range rather than a single figure. The commission distortions that motivate our analysis are small by comparison, once separated from the existence of the assistance channel itself.

7 Conclusion

This paper examines the welfare effects of decision assistance and commission-based steering in the ACA marketplace, combining a design-based analysis of decision assistance with a structural model of insurer pricing and broker steering, estimated on the population of California exchange enrollments from 2014 to 2019.

We find that decision assistance moves households toward silver plans and away from dominated choices, that private agents do so more effectively than public navigators, and that insurer commissions measurably steer broker-assisted enrollment. In the structural model, at the central valuation of the cost of being uninsured, the welfare gain from the guidance brokers provide, concentrated in improved risk protection, exceeds both the cost of the commissions that fund the channel and the premium spillover those commissions create for the rest of the market. The magnitude of that net gain, and whether it remains positive at the lowest valuation of coverage, depend on how heavily the health consequences of losing coverage are weighted, which is why we report it as a band rather than a point.

We close with three observations for policy. First, the case for restricting commissions rests on their differential component, the cross-plan variation that steers choices, which we estimate to be small, rather than on the broker channel itself, which we estimate to be valuable. Second, public and private assistance are not interchangeable. Navigators avoid the steering distortion but provide weaker guidance, so moving households from brokers to navigators improves the paternalistic navigator-rule measure while reducing objective welfare. Third, the direction of recent federal policy, the contraction of navigator funding, raises welfare in our model on both the revealed-preference and objective measures, though not on the navigator rule, so that the conclusion one draws depends on whether choices influenced by commissions are counted as welfare-relevant. We caution that the objective comparisons turn on a calibrated coefficient of risk aversion, and that the risk-protection channel driving them, while robust in sign, is the component of our welfare measure least tied to the data.

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Table 1: Summary Statistics for Covered California Enrollees[a]

Variable	Assisted	Unassisted	Overall
N	3,475,405	1,920,954	5,396,359
FPL	2.219	2.375	2.275
HH Size	1.463	1.338	1.419
% Male	0.456	0.483	0.465
% Age 0-17	0.022	0.022	0.022
% Age 18-34	0.261	0.372	0.300
% Age 55-64	0.325	0.244	0.296
% Black	0.020	0.030	0.024
% Hispanic	0.226	0.218	0.223
% Asian	0.176	0.128	0.159
% Dominated	0.022	0.039	0.027

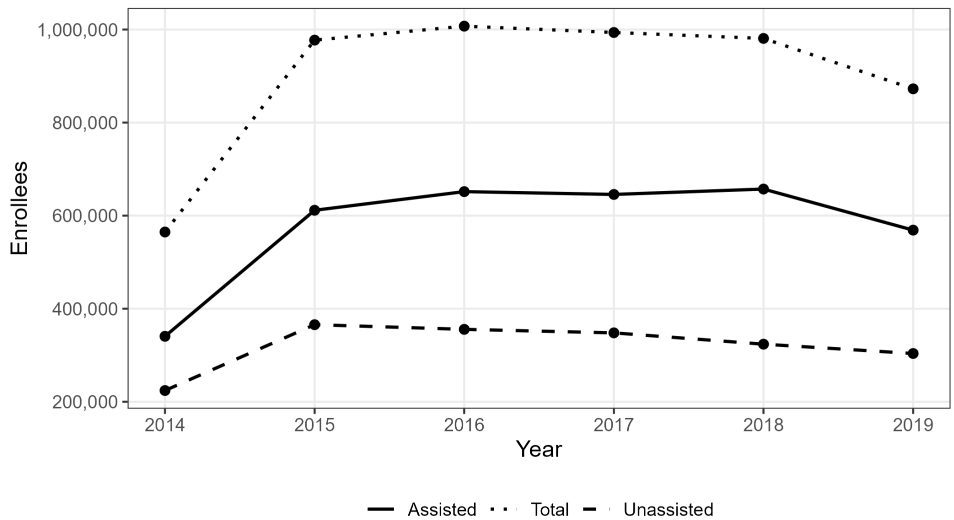
[a]Statistics based on the population of households enrolled in a Covered California plan from 2014–2019. Dominated choice rates are conditional on having at least one dominated plan in the choice set.

Tables and Figures

Figure 1: Commission Rates by Insurer, 2014–2019[a]

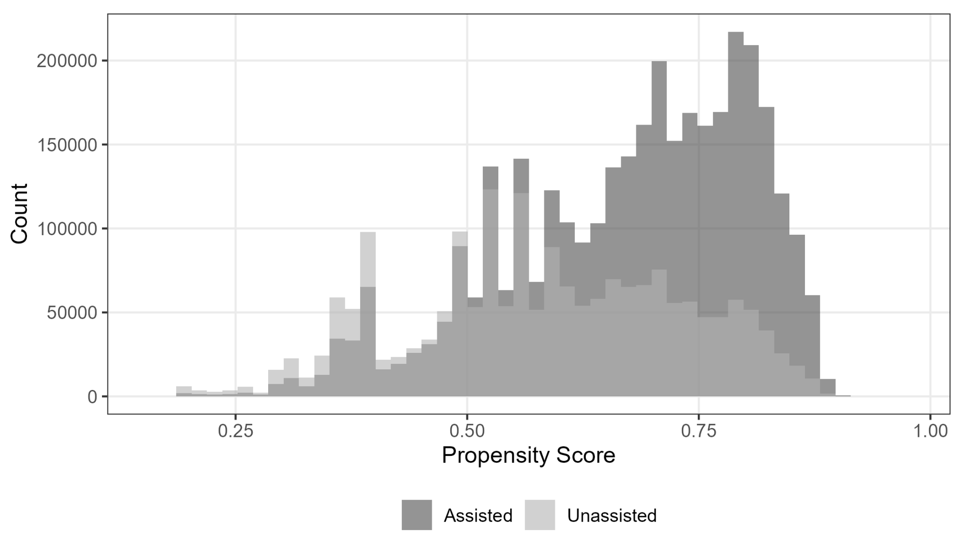
[a] Top panel shows flat-fee commissions (dollars per member per month). Bottom panel shows percentage-based commissions. Source: Covered California published rate schedules.

Figure 2: Enrollment by Channel, 2014–2019[a]



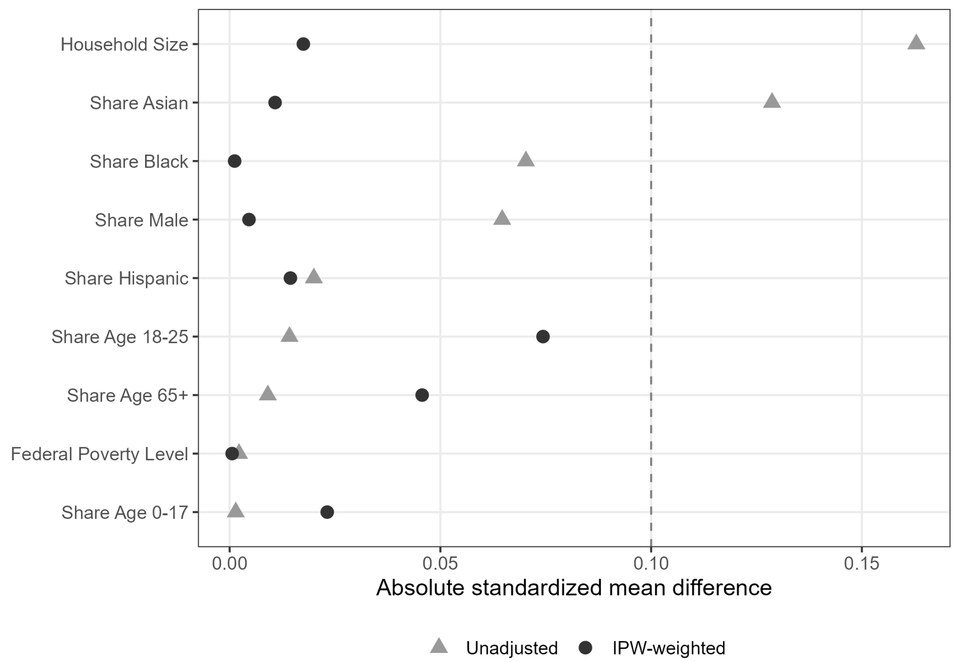
[a]Count of insured households enrolled in a Covered California plan in each year, by assistance channel.

Figure 3: Distribution of Propensity Scores for Decision Assistance[a]



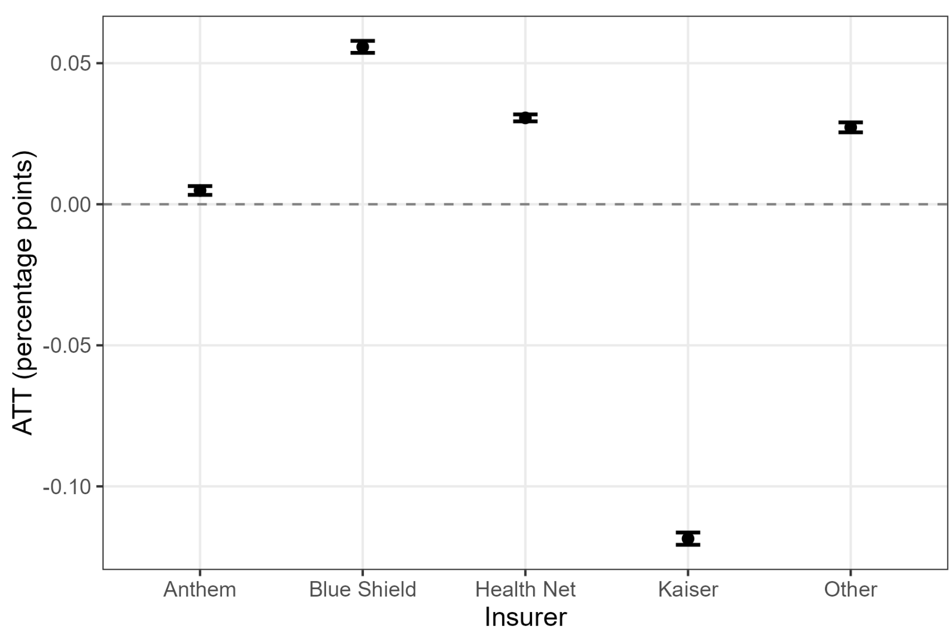
[a]Propensity scores from a logistic regression of decision assistance on household characteristics, including household income, racial and ethnic composition, age distribution, gender composition, and household size.

Figure 4: Covariate Balance Before and After Inverse-Propensity Weighting[a]



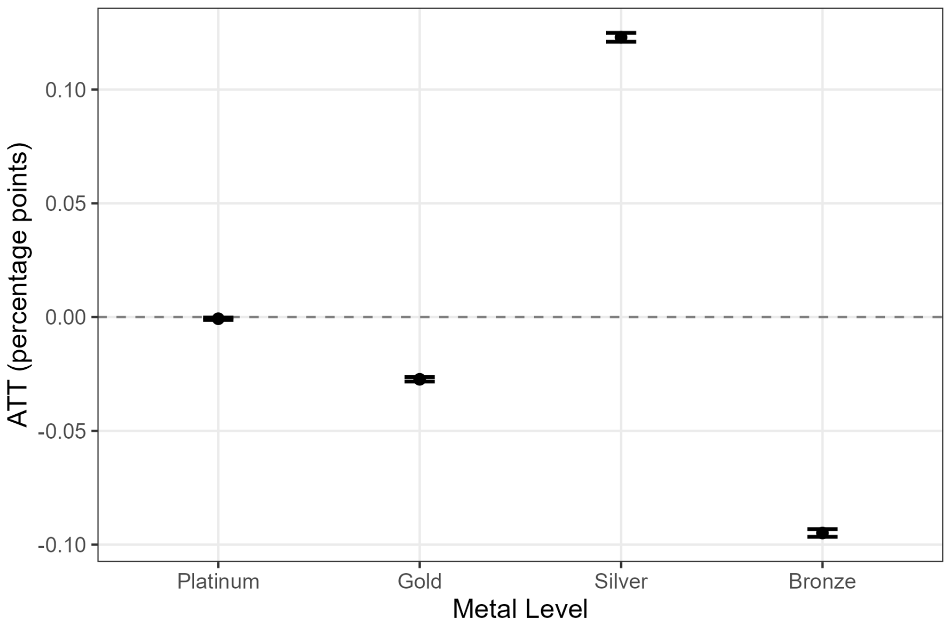
[a]Standardized mean differences between assisted and unassisted households. "Adjusted" reflects IPW-ATT weights where treated households receive weight 1 and untreated households receive weight $p/(1-p)$, with p denoting the estimated propensity score (see Figure 3).

Figure 5: ATT of Decision Assistance on Insurer Choice[a]



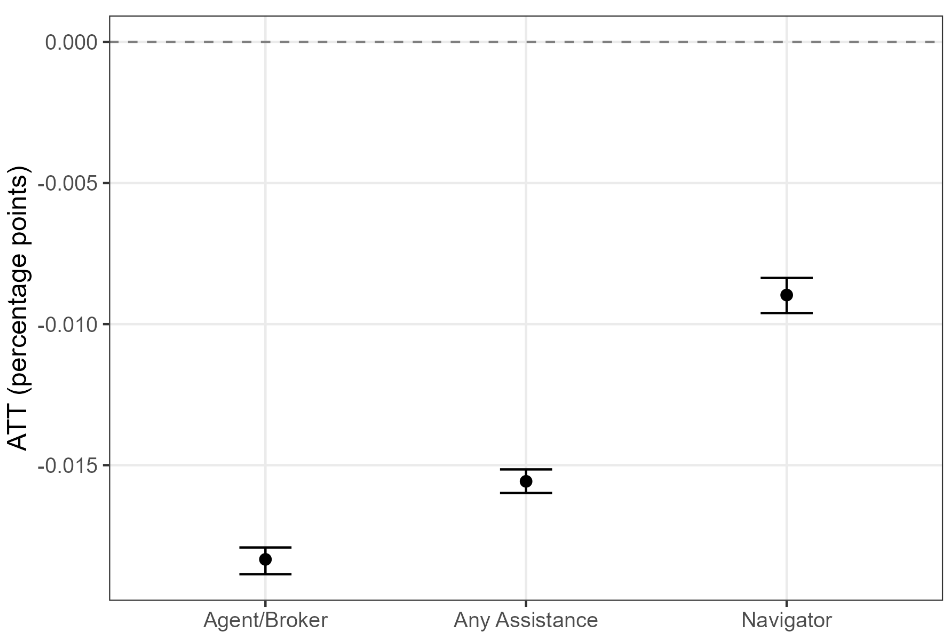
[a]Estimated ATT as described in Section 4, aggregated to the insurer level. Bars show 95% confidence intervals from a stratified bootstrap that resamples households within region-year cells.

Figure 6: ATT of Decision Assistance on Metal Tier Choice[a]



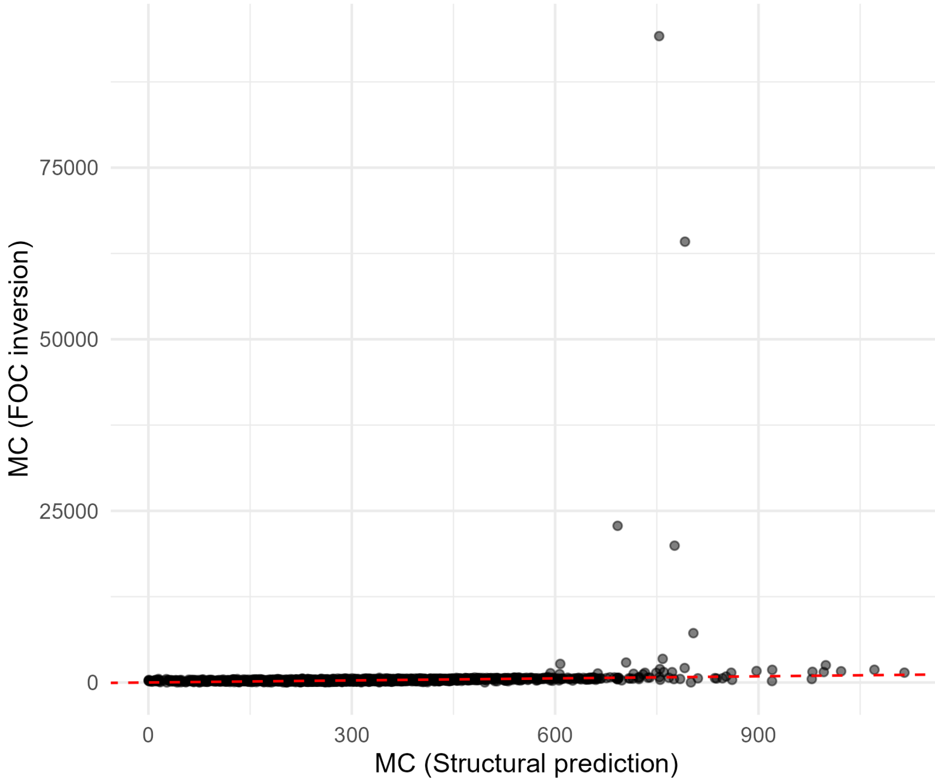
[a]Estimated ATT as described in Section 4, aggregated to the level of metal tier. Bars show 95% confidence intervals from a stratified bootstrap that resamples households within region-year cells.

Figure 7: ATT of Decision Assistance on Dominated Choices[a]



[a]Estimated ATT as described in Section 4.4, on the sample of households with at least one dominated plan in the choice set. Bars show 95% confidence intervals from a stratified bootstrap that resamples households within region-year cells.

Figure 8: Marginal Cost Validation[a]



[a]FOC-implied marginal cost versus structural prediction from risk score and claims regressions. Each point represents a plan in a region-year cell. The dashed line is the 45-degree line.

Table 2: Structural Demand Estimates[a]

Variable	Estimate	Std. Error
Premium	-0.6820	0.0045
Silver	0.4830	0.0053
Bronze	-1.6773	0.0104
HMO	-0.4974	0.0037
HSA	-0.7476	0.0046
Anthem	0.1848	0.0030
Blue Shield	0.3362	0.0028
Kaiser	0.7137	0.0028
Health Net	0.1542	0.0031
HH size × premium	-0.0283	0.0018
Age 0–17 × premium	0.2424	0.0106
Age 18–34 × premium	-0.3614	0.0034
Age 35–54 × premium	-0.2738	0.0029
Male × premium	-0.0242	0.0025
Black × premium	-0.1776	0.0066
Hispanic × premium	-0.3571	0.0032
Asian × premium	-0.3038	0.0036
Other race × premium	-0.2266	0.0027
FPL 250–400% × premium	0.0940	0.0027
FPL 400+% × premium	0.5190	0.0045
Age 0–17 × Silver	-1.0225	0.0195
Age 0–17 × Bronze	1.0931	0.0207
Age 18–34 × Silver	0.2759	0.0054
Age 18–34 × Bronze	0.6157	0.0066
Age 35–54 × Silver	0.0503	0.0050
Age 35–54 × Bronze	0.0087	0.0058
Male × Silver	-0.1080	0.0044
Male × Bronze	0.1054	0.0051
Navigator × Silver	1.3814	0.0090
Navigator × Bronze	1.8122	0.0117
Broker × Silver	1.4268	0.0071
Broker × Bronze	1.9259	0.0103
Navigator × premium	0.4618	0.0040
Broker × premium	0.4871	0.0033
Commission × broker	0.0206	0.0003
λ (nesting parameter)	0.6739	0.0024

[a]Estimated by maximum likelihood on the pooled sample of all households, weighted by household size, with household-clustered sandwich standard errors. Premiums are net of subsidy and measured in hundreds of dollars. The nesting parameter λ governs substitution between exchange plans and the outside option. Navigator × metal and broker × metal interactions capture each channel's effect on metal tier preferences. Commission × broker captures the effect of per-member-per-month commissions for broker-assisted households. Demographic × metal interactions allow the valuation of coverage generosity to vary with household age and gender composition.

Table 3: Cost Estimates[a]

Variable	Estimate
Risk score equation	
Constant	-1.4164
	(0.4221)
Actuarial value	3.8294
	(0.3042)
Share age 18–34	-1.1665
	(0.8369)
Share age 35–54	-2.0236
	(0.5098)
Share male	-0.5139
	(0.9683)
Anthem	0.0084
	(0.0477)
Blue Shield	-0.0738
	(0.0516)
Health Net	0.0098
	(0.0482)
Molina	-0.4037
	(0.0554)
L.A. Care	-0.1688
	(0.0963)
Sharp	-0.0765
	(0.1072)
Chinese Community	-0.4379
	(0.0558)
Oscar	-0.0908
	(0.0548)
Western	-0.0423
	(0.0604)
Valley	-0.0577
	(0.0878)
Claims equation	
Constant	5.5959
	(0.0774)
Log predicted risk score	1.1929
	(0.0909)
HMO	-0.1972
	(0.0487)
Linear trend	0.0769
	(0.0180)

[a]Two-step generalized method of moments estimates of the risk score equation (9) and claims equation (10), with the pricing first-order condition residuals included in the moment stack and the second-step weight equal to the inverse of the estimated moment covariance. Sandwich standard errors in parentheses. Kaiser is the omitted insurer in the risk score fixed effects. Demographic shares are predicted from the estimated choice model.

Table 4: Supply-Side Results by Metal Tier[a]

Metal	N	Premium	Markup	MC (FOC)	MC (Structural)	Lerner	Commission
Bronze	796	289.77	57.04	226.72	266.91	0.198	10.27
Silver	588	387.52	97.54	281.20	289.41	0.262	11.66
Gold	645	456.57	125.37	279.72	237.50	0.299	12.50
Platinum	588	529.22	138.09	359.33	333.47	0.280	14.00

[a]Each column reports the within-tier median across plan-cells. Markups recovered from the Bertrand-Nash pricing first-order conditions. MC (FOC) is the marginal cost implied by the first-order condition inversion. MC (Structural) is the predicted marginal cost from the risk score and claims equations estimated on rate filing data, net of risk adjustment transfers and reinsurance. Lerner index is markup divided by posted premium.

Table 5: Counterfactual Premium, Coverage, and Welfare Effects[a]

Scenario	τ	Δ Premium	Δ Uninsured (pp)	Δ CS	ΔV^{nav}	ΔV^{obj}
Observed	–	0.00	0.0	0.00	0	0
Zero commission	0.00	-144.32	7.8	-529.67	-881	-393
Zero commission	0.25	-124.25	6.4	-428.82	-647	-328
Zero commission	0.50	-86.23	4.8	-311.36	-397	-259
Zero commission	0.75	-63.45	3.3	-201.92	-275	-196
Zero commission	1.00	-13.62	0.9	-94.52	112	-103
Uniform commission	–	15.00	-0.1	9.05	18	8
Aligned commissions	–	11.83	-0.1	7.48	82	16
Scaled commission (25%)	–	-27.90	0.4	-50.87	39	-23
Scaled commission (50%)	–	-16.92	0.2	-34.23	24	-15
Scaled commission (75%)	–	-5.83	0.1	-17.20	8	-7
Navigator expansion	0.50	-30.33	0.4	-57.35	41	-20
Navigator expansion	1.00	-58.15	0.8	-97.67	62	-53
Flat-fee mandate	–	10.74	-0.0	4.42	-6	8
Navigator defunding (50%)	–	37.67	-0.4	49.48	-51	15
Navigator defunding (100%)	–	96.42	-1.0	125.55	-79	60

[a]Each row reports the change relative to the observed equilibrium, averaged across region-year cells, for a given counterfactual scenario. τ is the substitution rate for the zero-commission and navigator-expansion families and is blank where it does not apply. Δ Premium is the share-weighted change in posted premiums. Δ Uninsured is the change in the share of households taking the outside option, in percentage points. Δ CS is the change in Small-Rosen consumer surplus, ΔV^{nav} is the change in the navigator-rule money metric, and ΔV^{obj} is the change in the objective money metric at the central uninsured-cost calibration; all are defined in Section 6.3, and the four dollar figures (Δ Premium, Δ CS, ΔV^{nav}, and ΔV^{obj}) are per member per year. The low and high uninsured-cost cases are in Table 6.

Table 6: Objective Welfare Effect by Uninsured-Cost Calibration[a]

Scenario	ΔV^{obj} low	central	high
Observed	0	0	0
Zero commission			
$\tau = 0.00$	117	-393	-681
$\tau = 0.25$	95	-328	-567
$\tau = 0.50$	54	-259	-436
$\tau = 0.75$	23	-196	-319
$\tau = 1.00$	-44	-103	-137
Uniform commission	3	8	11
Aligned commissions	10	16	19
Scaled commission			
25%	1	-23	-36
50%	1	-15	-23
75%	1	-7	-11
Navigator expansion			
$\tau = 0.50$	6	-20	-35
$\tau = 1.00$	1	-53	-83
Flat-fee mandate	4	8	10
Navigator defunding			
50%	-8	15	28
100%	-7	60	97

[a]Change in the objective money-metric welfare measure relative to the observed equilibrium, per member per year, under the low, central, and high valuations of the cost of being uninsured described in Section 6.3. The band reflects the calibration of the cost of being uninsured, principally the mortality valuation; it is distinct from the sampling uncertainty in the estimated parameters. Scenarios are described in Section 6.1.

Table 7: Coverage and Welfare Effects with Bootstrap Standard Errors[a]

Scenario	Δ Uninsured (pp)	ΔV^{obj}
Remove assistance (τ=0)	7.8 (0.02)	-393 (1.6)
Brokers to navigators (τ=1)	0.9 (0.02)	-103 (1.7)
Uniform commission	-0.1 (0.00035)	8 (0.037)
Aligned commissions	-0.1 (0.0003)	16 (0.11)
Navigator expansion	0.8 (0.02)	-53 (1.5)
Flat-fee mandate	-0.0 (0.00027)	8 (0.074)
Navigator defunding	-1.0 (0.01)	60 (0.68)

[a]Change in the uninsured share (percentage points) and in the objective money metric at the central uninsured-cost calibration, per member per year, relative to the observed equilibrium. Standard errors in parentheses are from a parametric bootstrap over the estimated demand parameters and capture the sampling uncertainty in those estimates, distinct from the calibration band across uninsured-cost valuations in Table 6. Scenarios are described in Section 6.1.

Table 8: Producer Surplus and Government Cost by Scenario[a]

Scenario	τ	Δ Producer surplus	Δ Gov. subsidy
Observed	–	0	0
Zero commission	0.00	-196	-451
Zero commission	0.25	-157	-391
Zero commission	0.50	-115	-300
Zero commission	0.75	-64	-228
Zero commission	1.00	24	-97
Uniform commission	–	8	22
Aligned commissions	–	12	21
Scaled commission (25%)	–	6	-43
Scaled commission (50%)	–	6	-27
Scaled commission (75%)	–	5	-11
Navigator expansion	0.50	5	-45
Navigator expansion	1.00	-8	-100
Flat-fee mandate	–	13	16
Navigator defunding (50%)	–	4	49
Navigator defunding (100%)	–	9	147

[a]Change relative to the observed equilibrium, per member per year, averaged across region-year cells. Producer surplus is the insurer margin, posted premium less marginal cost, summed over enrolled members and net of commissions paid to brokers. Government subsidy is the advance premium tax credit actually paid, capped at the household premium, on enrolled households. Scenarios are described in Section 6.1.

Table 9: Coverage Effect: Welfare Decomposition and Sensitivity[a]

Panel A. Welfare change from removing assistance ($/member/year)	
Total	-647
Coverage margin	-725
Plan-choice margin	77
Panel B. Coverage effect (change in uninsured share, pp)	
Nesting parameter $\lambda = 0.50$	14
$\lambda = 0.67$ (estimated)	12
$\lambda = 0.85$	9
$\lambda = 1.00$	7
Outside-option value -1 util	7
baseline	12
+1 util	17

[a]Panel A decomposes the demand-side objective welfare change from removing decision assistance, holding premiums fixed, into a coverage margin (households leaving the market, valuing those who remain at their observed plan mix) and a plan-choice margin (reshuffling among those who remain insured), using an order-independent decomposition, in dollars per member per year. Panel B reports the coverage effect, the change in the uninsured share in percentage points, as the nesting parameter and the value of the outside option are varied around their estimated values. Both are demand-side predictions that hold premiums at their observed levels.

Figure 9: Premium Rating Regions in the California Exchange[a]

[a]California's 58 counties are divided into 19 rating areas. Source: California Department of Managed Health Care.